BMJ Open

Clinical prediction models for hospital falls: a scoping review protocol

Rex Parsons ,[1] Susanna M Cramb ,[1,2] Steven M McPhail[1,3]

To cite: Parsons R, Cramb SM, McPhail SM. Clinical prediction models for hospital falls: a scoping review protocol. *BMJ Open* 2021;**11**:e051047. doi:10.1136/bmjopen-2021-051047

► Prepublication history and additional supplemental material for this paper are available online. To view these files, please visit the journal online (http://dx.doi.org/10.1136/bmjopen-2021-051047).

Received 10 March 2021
Accepted 23 August 2021

Check for updates

© Author(s) (or their employer(s)) 2021. Re-use permitted under CC BY-NC. No commercial re-use. See rights and permissions. Published by BMJ.

[1]Australian Centre for Health Services Innovation and Centre for Healthcare Translation, School of Public Health and Social Work, Queensland University of Technology, Kelvin Grove, Queensland, Australia
[2]Jamieson Trauma Institute, Royal Brisbane and Women's Hospital, Metro North Health, Herston, Queensland, Australia
[3]Clinical Informatics Directorate, Metro South Health, Woolloongabba, Queensland, Australia

Correspondence to
Dr Rex Parsons;
rex.parsons@hdr.qut.edu.au

ABSTRACT

Introduction Falls remain one of the most prevalent adverse events in hospitals and are associated with substantial negative health impacts and costs. Approaches to assess patients' fall risk have been implemented in hospitals internationally, ranging from brief screening questions to multifactorial risk assessments and complex prediction models, despite a lack of clear evidence of effect in reducing falls in acute hospital environments. The increasing digitisation of hospital systems provides new opportunities to understand and predict falls using routinely recorded data, with potential to integrate fall prediction models into real-time or near-real-time computerised decision support for clinical teams seeking to mitigate fall risk. However, the use of non-traditional approaches to fall risk prediction, including machine learning using integrated electronic medical records, has not yet been reviewed relative to more traditional fall prediction models. This scoping review will summarise methodologies used to develop existing hospital fall prediction models, including reporting quality assessment.

Methods and analysis This scoping review will follow the Arksey and O'Malley framework and its recent advances, and will be reported using Preferred Reporting Items for Systematic Reviews and Meta-Analyses extension for Scoping Reviews recommendations. Four electronic databases (CINAHL via EBSCOhost, PubMed, IEEE Xplore and Embase) will be initially searched for studies up to 12 November 2020, and searches may be updated prior to final reporting. Additional studies will be identified by reference list review and citation analysis of included studies. No restriction will be placed on the date or language of identified studies. Screening of search results and extraction of data will be performed by two independent reviewers. Reporting quality will be assessed by the adherence to the Transparent Reporting of a multivariable prediction model for Individual Prognosis Or Diagnosis.

Ethics and dissemination Ethical approval is not required for this study. Findings will be disseminated through peer-reviewed publication and scientific conferences.

Strengths and limitations of this study

- ► The proposed review will be the first scoping review of hospital fall prediction model development processes.
- ► This protocol includes a systematic search and adheres to a rigorous, established research framework for study selection and reporting.
- ► The review will not include a meta-analysis of prediction performance.
- ► The review will not include studies relating to falls occurring out of the hospital setting.

INTRODUCTION

Falls are one of the most reported adverse events in hospitals internationally and are associated with significant disease burden and costs.[1] A 2015 audit of inpatient falls in the UK reported 6.63 falls per 1000 occupied bed days in acute care hospitals,[2] up from 4.8 in 2005–2006,[3] despite concerted efforts and substantial investment in patient safety measures. Falls have been reported to be associated with additional length of stay and operational costs internationally, including in the USA,[4] Canada[5] and Australia.[6 7]

Multifactorial assessments and associated interventions may reduce falls, suggesting that at least some proportion of hospital falls are preventable.[1 8 9] However, evidence for these approaches may be setting-dependent, with clinical trial evidence supporting only fall prevention approaches that incorporate fall risk assessment in subacute care hospital rehabilitation settings for older adults[10] but not in conventional acute care hospital ward settings.[11] This has caused some to question the value of current approaches to assessing fall risk and targeting fall prevention interventions in acute care hospital settings.[12] A recent non-inferiority trial across 10 hospitals indicated falls did not increase when staff ceased completing multifactorial fall risk assessments and instead used their own clinical reasoning to select interventions from a clinical decision support intervention list without completing the multifactorial fall risk assessment.[13] In addition to demonstrating non-inferiority, after adjusting for historical fall rates at the participating hospitals, the hospital fall incident rate ratio favoured the group that ceased fall risk assessment ratings.[13]

BMJ Open: first published as 10.1136/bmjopen-2021-051047 on 13 September 2021. Downloaded from http://bmjopen.bmj.com/ on March 10, 2026 by guest. Protected by copyright, including for uses related to text and data mining, AI training, and similar technologies.

The inability of conventional approaches to identifying patients at risk of falling, including fall risk screening questions and fall risk assessment tools, to successfully guide interventions to reduce fall injuries in acute care hospital settings may be due, at least in part, to poor fall prediction performance. Patient attributes that have been reported to influence fall risk include mental status,[14] toileting needs,[15] mobility impairment,[16] history of falls,[17] medications,[18] diabetes and poor blood glucose control,[19] and age,[20] among many others.[21] However, systematic reviews of conventional approaches to predict fall risk in hospitals report that their performance may not be considered clinically useful.[22] In response to this, the current guidelines from the National Institute for Health and Care Excellence (NICE) advise a pragmatic and simplistic approach to categorising patients as high fall risk if they are either (1) aged 65 years or older or (2) aged 50–64 years and judged by a clinician to be at higher risk of falling because of an underlying condition.[23] Since the NICE guidelines classify most patients in hospital wards as being at high risk of a fall event which occurs for approximately 3.6% of patients,[6] this approach may be sensitive but lacks specificity for guiding specific fall prevention interventions.

The increasing digitisation of hospital information systems in recent years, including the wider adoption of integrated electronic medical records (ieMRs), has laid a foundation for the development and adoption of more advanced approaches to fall risk prediction in hospitals. The use of additional predictors, obtained from ieMR and other digital hospital systems, has been shown to improve risk prediction performance in related clinical contexts.[24] In addition, the potential to integrate high-performing prediction models nested within these systems may enable continuous risk predictions and computerised clinical decision support that is potentially desirable to clinicians[25] and medical associations.[26] The wide array of fields routinely and automatically recorded in ieMRs and associated systems provides an opportunity for the application of a range of advanced statistical modelling and algorithmic approaches that have potential utility in hospital settings.[27] However, studies reporting newer approaches to fall risk prediction, including machine learning and its potential role in computerised clinical decision support systems intended to reduce fall risk, have not yet been systematically collated and considered alongside more conventional methods for predicting hospital falls. The quality of reporting for inpatient fall prediction models has also not been described previously. It is also unknown whether the publication of the Transparent Reporting of a multivariable prediction model for Individual Prognosis Or Diagnosis (TRIPOD) statement[28] has led to improved reporting quality in the context of inpatient fall prediction models.

Study rationale

Although a range of approaches to assessing fall risk in hospitals have been available for decades,[29] performance is inconsistent,[30] and validation studies suggest that generalisability of some approaches to fall risk prediction may be poor.[31] Increasing completion rates of conventional approaches to multifactorial fall risk assessment and implementation of associated interventions do not have clinical trial evidence of effect in acute care hospital wards.[32] Poor performance of conventional and non-conventional approaches to fall risk prediction in hospital environments may be influenced by the methods used to develop these fall risk prediction approaches. This scoping review will allow us to summarise methods and sources of data which have already been used in the development of published approaches to in-hospital fall risk prediction, and potentially identify underexplored methods and data sources. It is anticipated that these findings will provide insight into promising approaches for improving fall risk prediction models that have potential to be adopted in computerised decision support solutions suitable for hospital settings, including ieMR environments. Therefore, this review will aim to describe existing approaches to hospital fall prediction model development and the quality of reporting in these studies.

Study objectives

To address this aim, the following objectives have been set: (1) describe sources and predictor variables used for model development in the context of in-hospital fall prediction, (2) describe the development process and algorithmic approaches used, and (3) describe how existing in-hospital fall prediction model reporting adheres to the TRIPOD guidelines.[28]

METHODS

This scoping review will be reported to the Preferred Reporting Items for Systematic reviews and Meta-Analyses extension for Scoping Reviews.[33] It will be guided by the Arksey and O'Malley framework[34] as well as recent advances.[35] The framework includes the following five steps:

1. Identifying the research question.
2. Identifying relevant studies.
3. Selecting studies.
4. Charting the data.
5. Collating, summarising and reporting the results.

This review design may identify areas where a systematic review or a meta-analysis is desired to answer a more precise and quantitative research question, but no meta-analysis is planned for inclusion in this scoping review.

Stage 1: identifying the research question

The purpose of a scoping review is to map the literature within a given field, in this case, hospital fall prediction models. Therefore, we will use an iterative process as our understanding of the literature improves, and the questions may be reframed appropriately.

Our review will focus on the following research questions:

BMJ Open: first published as 10.1136/bmjopen-2021-051047 on 13 September 2021. Downloaded from http://bmjopen.bmj.com/ on March 10, 2026 by guest. Protected by copyright, including for uses related to text and data mining, AI training, and similar technologies.

1. What clinical prediction models are available for hospital falls?
2. What methods are used to create these models?
3. What predictor variables are used in these models?
4. How well have existing models been reported?

Stage 2: identifying relevant studies

Studies that match the search terms provided will be extracted from CINAHL via EBSCOhost, PubMed, IEEE Xplore and Embase, and will include studies available from database inception to 12 November 2020. This search may be updated prior to submission of the manuscript reporting the findings from this scoping review to ensure the most recent literature is included in the review at time of publication. There will not be any limitations on the document type. The search terms we will use for each database were selected with the consultation of a librarian and can be found in the online supplemental appendix 1.

Eligibility criteria

To enable the exploration of processes and data used to develop approaches for fall risk prediction among hospital inpatients, this review will include studies that developed fall risk prediction approaches and reported a measure of predictive model performance in their development study. Studies that did not report a comparable measure of predictive performance for in-hospital fall prediction will be excluded. Similarly, studies that validated existing models in new samples, focused on use of sensors, accelerometers or other equipment not likely to be widely available in hospitals, or predicted falls that occur within a clinically irrelevant period will be excluded. For the purpose of this review, we consider a clinically irrelevant period to be one that included falls that occurred after hospitalisation, or within the hospitalisation period but with insufficient time to realistically enable a clinical team to be notified of the fall risk and implement risk mitigating actions (eg, studies predicting falls that occur within one minute).

Stage 3: study selection

Search results from each database will be imported into EndNote X9 and duplicates will be removed before being imported into Rayyan.[36] Article titles and abstracts will be assessed against the eligibility criteria by two reviewers. Conflicts will be resolved by discussion or by an additional reviewer.

The same researchers will review the full texts of the included studies against the same eligibility criteria. After full-text review, the reference lists of included studies will be examined for additional studies of relevance.

Stage 4: charting the data

Two reviewers will extract data from 20% of included studies, independently, and discrepancies will be discussed. One of the reviewers will then complete data extraction for all remaining studies. The fields for data extraction adapted from the Joanna Briggs Institute template[37] will include those described in the online supplemental appendix 1. Data fields to be extracted were chosen based on the TRIPOD statement[28] and include identifying details of the study, its design, sample characteristics (including sample size and patient age), data sources, type of fall outcome being predicted, modelling approach and performance.

Stage 5: collating, summarising and reporting the results

A narrative report will summarise the studies with regard to the research questions. Tables will be used to present detailed descriptions of each included study with regard to (1) study design and data sources; (2) model development process, features used and validation method(s); and (3) adherence to TRIPOD reporting guidelines.[28] Aggregated results will also be presented in tables to provide the end user with a summarised view of approaches and strategies used. These results will be interpreted with regard to the research aims and objectives. Gaps in the literature will be identified concerning current standards of reporting and development methods for inpatient fall prediction models.

Patient and public involvement

No patients or members of the public were involved in this study, and the study will not specifically include additional stakeholder engagement as the methods and processes being reviewed are already well established.

Contributors Design of the protocol—RP, SC and SMM. Draft of the manuscript—RP. Review and final approval of the manuscript—RP, SC and SMM.

Funding This work was supported by funding from the Digital Health Cooperative Research Centre (DHCRC-0058) and SMM is supported by an National Health and Medical Research Council administered fellowship (#1181138). The funders had no role in study design or decision to submit for publication.

Competing interests All authors have completed the Unified Competing Interest form (available on request from the corresponding author) and declare no support from any organisation for the submitted work, no financial relationships with any organisations that might have an interest in the submitted work in the previous 3 years and no other relationships or activities that could appear to have influenced the submitted work.

Patient consent for publication Not required.

Provenance and peer review Not commissioned; externally peer reviewed.

Supplemental material This content has been supplied by the author(s). It has not been vetted by BMJ Publishing Group Limited (BMJ) and may not have been peer-reviewed. Any opinions or recommendations discussed are solely those of the author(s) and are not endorsed by BMJ. BMJ disclaims all liability and responsibility arising from any reliance placed on the content. Where the content includes any translated material, BMJ does not warrant the accuracy and reliability of the translations (including but not limited to local regulations, clinical guidelines, terminology, drug names and drug dosages), and is not responsible for any error and/or omissions arising from translation and adaptation or otherwise.

Open access This is an open access article distributed in accordance with the Creative Commons Attribution Non Commercial (CC BY-NC 4.0) license, which permits others to distribute, remix, adapt, build upon this work non-commercially, and license their derivative works on different terms, provided the original work is properly cited, appropriate credit is given, any changes made indicated, and the use is non-commercial. See: http://creativecommons.org/licenses/by-nc/4.0/.

ORCID iDs
Rex Parsons http://orcid.org/0000-0002-6053-8174
Susanna M Cramb http://orcid.org/0000-0001-9041-9531

BMJ Open: first published as 10.1136/bmjopen-2021-051047 on 13 September 2021. Downloaded from http://bmjopen.bmj.com/ on March 10, 2026 by guest. Protected by copyright, including for uses related to text and data mining, AI training, and similar technologies.

REFERENCES

1 Cameron ID, Dyer SM, Panagoda CE, *et al*. Interventions for preventing falls in older people in care facilities and hospitals. *Cochrane Database Syst Rev* 2018;9:CD005465.
2 Royal College of physicians. National audit of inpatient falls: audit report. London; 2015.
3 National Patient Safety Agency. *Slips, trips and falls in hospitals*, 2007. http://www.mtpinnacle.com/pdfs/slips-trips-fall-2007.pdf
4 Wong CA, Recktenwald AJ, Jones ML, *et al*. The cost of serious fall-related injuries at three midwestern hospitals. *Jt Comm J Qual Patient Saf* 2011;37:81–7.
5 Zecevic AA, Chesworth BM, Zaric GS, *et al*. Estimating the cost of serious injurious falls in a Canadian acute care hospital. *Can J Aging* 2012;31:139–47.
6 Morello RT, Barker AL, Watts JJ, *et al*. The extra resource burden of in-hospital falls: a cost of falls study. *Med J Aust* 2015;203:367.
7 Hill KD, Vu M, Walsh W. Falls in the acute hospital setting--impact on resource utilisation. *Aust Health Rev* 2007;31:471–7.
8 Oliver D, Connelly JB, Victor CR, *et al*. Strategies to prevent falls and fractures in hospitals and care homes and effect of cognitive impairment: systematic review and meta-analyses. *BMJ* 2007;334:82–5.
9 Miake-Lye IM, Hempel S, Ganz DA, *et al*. Inpatient fall prevention programs as a patient safety strategy: a systematic review. *Ann Intern Med* 2013;158:390–6.
10 Haines TP, Bennell KL, Osborne RH, *et al*. Effectiveness of targeted falls prevention programme in subacute hospital setting: randomised controlled trial. *BMJ* 2004;328:676.
11 Avanecean D, Calliste D, Contreras T, *et al*. Effectiveness of patient-centered interventions on falls in the acute care setting compared to usual care: a systematic review. *JBI Database System Rev Implement Rep* 2017;15:3006–48.
12 McPhail SM. *Are we allocating Hospital resources to fall prevention interventions that are completely ineffective?* 2016.
13 Morris ME, Haines T, Hill AM, *et al*. Divesting from a scored Hospital fall risk assessment tool (frat): a cluster randomized non-inferiority trial. *J Am Geriatr Soc* 2021;9. doi:10.1111/jgs.17125. [Epub ahead of print: 09 Apr 2021].
14 Atlas A, Kerse N, Rolleston A, *et al*. Falls and depression in octogenarians - life and living in advanced age: a cohort study in New Zealand. *J Prim Health Care* 2017;9:311–5.
15 Rose G, Decalf V, Everaert K, *et al*. Toileting-related falls at night in hospitalised patients: the role of nocturia. *Australas J Ageing* 2020;39:e70–6.
16 Kim J, Kim S, Park J, *et al*. Multilevel factors influencing falls of patients in hospital: the impact of nurse staffing. *J Nurs Manag* 2019;27:1011–9.
17 Butler Forslund E, Jørgensen V, Skavberg Roaldsen K, *et al*. Predictors of falls in persons with spinal cord injury—a prospective study using the downton fall risk index and a single question of previous falls. *Spinal Cord* 2019;57:91–9.
18 O'Neil CA, Krauss MJ, Bettale J, *et al*. Medications and patient characteristics associated with falling in the hospital. *J Patient Saf* 2018;14:27–33.
19 Yang Y, Hu X, Zhang Q, *et al*. Diabetes mellitus and risk of falls in older adults: a systematic review and meta-analysis. *Age Ageing* 2016;45:761–7.
20 Anderson DC, Postler TS, Dam T-T. Epidemiology of hospital system patient falls: a retrospective analysis. *Am J Med Qual* 2016;31:423–8.
21 Evans D, Hodgkinson B, Lambert L, *et al*. Falls risk factors in the hospital setting: a systematic review. *Int J Nurs Pract* 2001;7:38–45.
22 Matarese M, Ivziku D, Bartolozzi F, *et al*. Systematic review of fall risk screening tools for older patients in acute hospitals. *J Adv Nurs* 2015;71:1198–209.
23 National Institute for Health Care Excellence. *Falls in older people: assessing risk and prevention*. NICE, 2013.
24 Desai RJ, Wang SV, Vaduganathan M, *et al*. Comparison of machine learning methods with traditional models for use of administrative claims with electronic medical records to predict heart failure outcomes. *JAMA Netw Open* 2020;3:e1918962–e62.
25 Voruganti T, O'Brien MA, Straus SE, *et al*. Primary care physicians' perspectives on computer-based health risk assessment tools for chronic diseases: a mixed methods study. *J Innov Health Inform* 2015;22:333–9.
26 Maddox TM, Albert NM, Borden WB, *et al*. The learning healthcare system and cardiovascular care: a scientific statement from the American heart association. *Circulation* 2017;135:e826–57.
27 Nakatani H, Nakao M, Uchiyama H, *et al*. Predicting inpatient falls using natural language processing of nursing records obtained from Japanese electronic medical records: case-control study. *JMIR Med Inform* 2020;8:e16970.
28 Collins GS, Reitsma JB, Altman DG, *et al*. Transparent reporting of a multivariable prediction model for individual prognosis or diagnosis (TRIPOD): the TRIPOD statement. The TRIPOD group. *Circulation* 2015;131:211–9.
29 Oliver D, Britton M, Seed P, *et al*. Development and evaluation of evidence based risk assessment tool (stratify) to predict which elderly inpatients will fall: case-control and cohort studies. *BMJ* 1997;315:1049–53.
30 Park S-H. Tools for assessing fall risk in the elderly: a systematic review and meta-analysis. *Aging Clin Exp Res* 2018;30:1–16.
31 Oliver D, Papaioannou A, Giangregorio L, *et al*. A systematic review and meta-analysis of studies using the stratify tool for prediction of falls in hospital patients: how well does it work? *Age Ageing* 2008;37:621–7.
32 Barker AL, Morello RT, Wolfe R, *et al*. 6-PACK programme to decrease fall injuries in acute hospitals: cluster randomised controlled trial. *BMJ* 2016;352:h6781.
33 Tricco AC, Lillie E, Zarin W, *et al*. PRISMA extension for scoping reviews (PRISMA-ScR): checklist and explanation. *Ann Intern Med* 2018;169:467–73.
34 Arksey H, O'Malley L. Scoping studies: towards a methodological framework. *Int J Soc Res Methodol* 2005;8:19–32.
35 Pham MT, Rajić A, Greig JD, *et al*. A scoping review of scoping reviews: advancing the approach and enhancing the consistency. *Res Synth Methods* 2014;5:371–85.
36 Ouzzani M, Hammady H, Fedorowicz Z, *et al*. Rayyan-a web and mobile APP for systematic reviews. *Syst Rev* 2016;5:210.
37 Aromataris E, Munn Z. JBI manual for evidence synthesis. *JBI* 2020.

BMJ Open: first published as 10.1136/bmjopen-2021-051047 on 13 September 2021. Downloaded from http://bmjopen.bmj.com/ on March 10, 2026 by guest. Protected by copyright, including for uses related to text and data mining, AI training, and similar technologies.